

	Baylor University Statistical Sciences Department Waco, TX 76798 ☎: (803) 944-7265 ✉: Eva_Murphy@baylor.edu 🌐: https://evamurphy100.github.io/
CITIZENSHIP	United States
RESEARCH INTERESTS	Environmental science; Public Health; Spatio-Temporal statistics; Latent variable models; Causal inference.
EMPLOYMENT	Baylor University, Waco, TX - Assistant ProfessorAugust 2026 - Present Wake Forest University, Winston-Salem, N.C. - Teacher-Scholar Postdoctoral FellowJuly 2023 - June 2026 Mentored by Dr. Staci Hepler (primary mentor) and Dr. David Kline. Conduct independent and collaborative research in Bayesian spatial factor modeling and public health applications. Teach undergraduate courses including Introduction to Regression and Data Science and Elementary Probability and Statistics. Clemson University, Clemson, S.C. - Graduate Research AssistantSeptember 2022 - June 2023 - Graduate Teacher of RecordAugust 2018 - June 2022 Mentored by Dr. Whitney Huang. Conducted dissertation research on statistical modeling of wind speed and wind direction, developing joint, spatio-temporal, and directional extreme value models. Taught undergraduate courses including Pre-Calculus, Introductory Differential Calculus, Business Calculus I & II, and Calculus of a Single Variable.
EDUCATION	Ph.D. in Mathematical and Statistical Sciences, Clemson University, Clemson, SC, August 2023 Dissertation: A Framework for Statistical Modeling of Wind Speed and Wind Direction[Dissertation Link] Advisor: Dr. Whitney Huang, Associate Professor, School of Mathematical and Statistical Sciences Committee Members: Dr. Andrew D. Brown, Dr. Colin Gallagher, and Dr. Brook Russell M.S. in Mathematical Sciences, University of West Florida, Pensacola, FL, April 2018 - Advisor: Dr. Josaphat Uvah, Professor, University of West Florida B.S. in Mathematical Sciences, Babes-Bolyai University, Romania, July 2006

PUBLICATIONS

1. **Murphy E.**, Cademartori C., Kline D.M., Hurley R., Yong J., McDonnell C. L., Hepler S.A., Adams M.C.B, “Impact of Hurricane Florence on buprenorphine transactions for opioid use disorder: A dual-perspective synthetic control study.” The Journal of Climate Change and Health, Volume 29, 2026, 100679,ISSN 2667-2782, <https://doi.org/10.1016/j.joclim.2026.100679>.
2. **Murphy E.**, Kline D. M.,Hepler S. A. “The role of spatially varying loadings in dynamic spatial factor models for modeling the opioid syndemic.” Health Services and Outcomes Research Methodology 26.1 (2026): 71-86.
3. **Murphy E.**, Kline D. M., McKnight E., Bonny B., Waller, L., Miller, W. C., Hepler S. A. “Integrating data at multiple spatial scales to estimate the local burden of the opioid syndemic.” Spat Spatiotemporal Epidemiol. 2025 Jun;53:100720.
4. **Murphy E.**, Huang W. K., Bessac, J., Wang, J., Kotamarthi, R. “Joint modeling of wind speed and wind direction through a conditional approach.” Environmetrics, Volume 36, Issue 3, 2025 (Winner of ASA Section on Statistics and the Environment Student Paper Competition).
5. **Murphy E.**, Kline D. M., Egan, K. L., Lancaster, K. E., Waller, L., Miller, W. C., Hepler S. A. “Exploring the Opioid Syndemic in North Carolina: A Novel Approach to Modeling and Identifying Factors.” Biostatistics, Volume 26, Issue 1, 2025.
6. Kline, D. M., White, B. N., Lancaster, K. E., Egan, K. L, **Murphy, E.**, Miller, W. C, Hepler, S. A. “Estimating prevalence of opioid misuse in North Carolina counties from 2016-2021: An integrated abundance model approach.” Epidemiology. 2025 May 1;36(3):310-318.

MANUSCRIPTS IN PREPARATION

1. **Murphy, E**, Cademartori, C., Kline, D., Hurley, R., Yong, J., Hepler, S., Adams, M.C.B. “Impact of Resource Accessibility Disruptions on Pain Treatments.”
2. Enderle, M. N., **Murphy, E.**, Murphy, J. , Nwaozuru, U. C., Kline,D. M., and Lancaster, K. E. Willingness to receive long-acting injectable pre-exposure prophylaxis among people who inject drugs in Appalachian Ohio: An application of the information-motivation-behavioral skills model.
3. **Murphy E.**, Huang W.K., Brook Russell “ Modeling Directional Extreme Wind Speeds Using an Extreme Value Regression Approach”.
4. **Murphy E.**, Huang W.K., Ting Fung Ma “ Spatio-temporal modeling of wind speed data”.

AWARDS

Student Award

- **Outstanding Graduate in Research Award** 2023
This award is given by the School of Mathematical and Statistical Sciences, Clemson University.
- **ASA ENVR Student Paper Award** 2023
The award is sponsored by the Section on Statistics and the Environment (ENVR) of the American Statistical Association (ASA). Papers were judged on environmental importance/relevance, statistical modeling/novelty, as well as writing (including organization, clarity, and accessibility).
- **Dr. Kenyon Fairey Graduate Fellowship**, Clemson University
September 2022 - May 2023
“The fellowship’s purpose is to provide graduate fellowships for doctoral students whose scholarship and/or applied research focuses on resource management, conservation, outdoor education, or health and wellness. Students are selected through a competitive review process managed by the Graduate School.”

- **Call Me Doctor Dissertation Completion Fellowship**, Clemson University
August 2022 - August 2023

“The Call Me Doctor Fellowship offers a unique opportunity for students from underrepresented and diverse backgrounds to pursue their doctoral studies with financial support, professional development, and mentoring from faculty with interdisciplinary expertise.”

Presentation Competition

- **Poster Award**, SCROS Summer Research Conference, June 2024.
- **Best Talk Award**, SC-ASA Palmetto Symposium April 2022

Travel Awards

- Provost’s Fund for Faculty Travel, Wake Forest University October 2025
- Provost’s Fund for Faculty Travel, Wake Forest University October 2024
- SRC ASA/Kutner Faculty Poster Session Award June 2024
- Faculty Development Fund for Visiting Professors, Wake Forest University June and October 2024
- Joint Statistical Meeting - Student Paper Award Presentation, ENVR August 2023
- Machine Learning for Climate and Weather Applications, Institute for Mathematical and Statistical Innovations (IMSI) November 2022
- Detection and Attribution of Climate Change, IMSI October 2022
- ENVR 2022 Workshop, Provo, UT October 2022
- Climate Model Evaluation and Uncertainty, IMSI September 2022
- Graduate Student Travel, Clemson University August 2022

PRESENTATIONS

Invited Talks

Integrating data at multiple spatial scales to estimate the local burden of the opioid syndemic.

- Women in Statistics and Data Science Conference November, 2025
- International Day of Women in Statistics and Data Science, October, 2025

Understanding the Opioid Syndemic in North Carolina: A Novel Approach to Modeling and Identifying Factors

- ICHPS 2025 Conference January, 2025
- NESS 2025 June, 2025

Understanding the Opioid Syndemic using Factor Analysis

- Wake Forest Biostatistics and Data Science Seminar Series March, 2025

Spatio-temporal modeling of wind speed data

- AISC 2024 Conference October, 2024
- ICSA Applied Statistics Symposium June, 2023

Modeling Directional Extreme Wind Speed Using an Extreme Value Regression Approach

- Statistical Sciences Colloquium, Wake Forest University September, 2023

- Statmospheric Sciences Seminar, Berkeley National Laboratory, October 11, 2023

Joint modeling of wind speed and wind direction through a conditional approach

- Joint Statistical Meeting August, 2023
- 5th International Conference on Econometrics and Statistics, (virtual) June 2022
- Math For All Satellite Conference at Clemson University, SC Feb. 2022

Contributed Talks

Spatio-temporal modeling of wind speed data

- SC-ASA Palmetto Symposium, Columbia, SC April 2023

Joint modeling of wind speed and wind direction through a conditional approach

- Joint Statistical Meeting, Washington, DC August 2022
- Climate Informatics (virtual) May 2022
- SC-ASA Palmetto Symposium - Best Talk Award, Columbia, SC April 2022

Statistical framework for studying the spatio-temporal variation of wind speed and wind direction

- The Graduate Student Network Conference, NISS, (virtual) June 2021

Contributed Posters

Spatio-temporal modeling of wind speed data

- ENVR 2024 Workshop, Boulder, CO October, 2024

Understanding the Opioid Syndemic in North Carolina: A Novel Approach to Modeling and Identifying Factors

- SRCOS 2024 Summer Research Conference, Clemson, SC June, 2024

Joint modeling of wind speed and wind direction through a conditional approach

- Climate Model Evaluation and Uncertainty, Chicago, IL September 2022
- ENVR Workshop, Provo, UT October, 2022

Modeling of Wind Speed and Wind Direction

- The Graduate Student Network Conference, NISS, (virtual) May 2022

General Audience

On the road of modeling wind speed and wind direction

- Math Club, Clemson University March 2023

Modeling wind speed and wind direction

- Clemson Calculus Challenge, Clemson University April, 2023

TEACHING
EXPERIENCE

Teacher-Scholar Postdoctoral Fellow, Wake Forest University

- STA 111: Elementary Probability and Statistics (4 sections over 2 semesters).
- STA 112: Introduction to Data Science and Regression (6 sections over 3 semesters).

Graduate Teacher of Record, School of Mathematical and Statistical Sciences, Clemson University

- Business Calculus I (2 sections over 2 semesters).
- Business Calculus II (2 sections over 4 semesters) .
- Pre-Calculus and Introductory Differential Calculus (2 sections over 1 semester).
- Calculus of Single Variable (2 sections over 1 semester).

Adjunct Instructor of Mathematics, Piedmont Technical College, Newberry, SC and Midlands Technical College, Columbia, SC

- Probability and Statistics (1 online section over 4 semesters).
- Beginning Algebra (1 section over 2 semesters).
- Intermediate Algebra (1 section 1 semester).
- Algebra, Geometry and Trigonometry (1 section 1 semester).

Teacher of Mathematics, Mid-Carolina High School, Newberry, SC

- Subjects taught: Algebra 1, Algebra 2, Geometry, Pre-Calculus, AP Calculus AB.

SERVICE

Session Chair

- The Statistical Analyzes for Environmental Monitoring, Contributed Papers Session, Section on Statistics and the Environment, Joint Statistical Meeting, August 2022

Paper Reviewer for:

- *Weather and Climate Extremes*
- *Advances in Statistical Climatology, Meteorology and Oceanography*
- *International Journal of Drug Policy*
- *Journal of Urban Health*
- *Journal of Agricultural, Biological and Environmental Statistics*
- *International Journal of Drug Policy*
- *Data Science in Science*
- 11th *International Conference on Climate Informatics* - extended abstract review

Mentor:

- STRIVE for MORE conference, (virtual) September 2020 and 2021

HARD SKILLS

R, MATLAB, Python, L^AT_EX, Unix/Linux Shell Scripting, High Performance Computing (HPC)

PROFESSIONAL
MEMBERSHIP

American Statistical Association (ASA)	Since August. 2018
ENVR Section. ASA	Since August 2018
Institute of Mathematical Statistics (IMS)	Since October 2022